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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental Models: Cell Lines			
Materials and Methods Cell Lines and Cell Culture In our study, we used four different cell lines that were chosen in order to consider features and cellular diversity of different organs of origin (33): NIH-3T3 mouse fibroblasts (connective tissue), NRK-52E rat epithelial cells (kidney), RLE-6TN alveolar epithelial cells (lung), and RAW 264.7 mouse macrophages (immune system).	RAW 264.7		Suggestion: RRID:CVCL_0493(link)
NIH-3T3 mouse embryonic fibroblasts, obtained by ATCC (No. ATCC® CRL-1658TM), were cultured in DMEM supplemented with 10% FCS, 1 mM pyruvate, and 2 mM glutamine, passaged three times a week.	NIH-3T3 mouse embryonic fibroblasts		
For the experiments with fixation, cells were harvested and seeded into Tomodishes (Tomocube, Inc., Daejeon, Republic of Korea) from the culture and cultivated overnight (for RAW 264.7-30,000 cells per Petri dish, for NIH-3T3, NRK-52E, RLE-6TN – 10,000 cells per Petri dish in 800 µl of medium).	RAW 264.7-30,000		
	NIH-3T3		
– Mirrors , DMD - digital micromirror device (adapted from reference (25)); (b) 3D RI distribution visualization; (c) refractive index distribution on x-y , x-z , and y-z cross sections in live cell from NIH3T3 line.	NIH3T3		Suggestion: RRID:CVCL_0594(link)
Software and Algorithms			
The local segmentation of internal organelles (nucleus, nucleolus, nucleoplasm, and cytoplasm) in the RI images was performed manually by using a MATLAB R2018b script.	MATLAB		Suggestion: MATLAB, RRID:SCR_001622(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Relative changes of the RI difference between nucleolus and nucleus, nucleolus and cytoplasm were determined and the changes of this difference before and after PFA fixation were calculated and the significance of the differences was assessed by the Student's t test, or by the Mann–Whitney U test, if the assumption of normality was not met (which was evaluated by Shapiro–Wilk test).	Student's t test
	Mann–Whitney U test
	Shapiro–Wilk test

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq